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THE PALAEOMAGNETISM OF THE MAIN ZONE OF THE EASTERN BUSHVELD COMPLEX

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ABSTRACT

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Palaeomagnetic data were acquired from eighteen sampling sites situated in the main zone of the eastern Bushveld Complex, Transvaal, South Africa. Specimens were subjected to alternating field and thermal demagnetization. Two mean magnetization directions, which are approximately antipodal, were found. One direction represents subzone B of the main zone in the eastern Bushveld Complex and yields a palaeomagnetic pole at 17.3°/35.7°E. The second direction represents subzone C of the main zone in the eastern Bushveld Complex with virtual geomagnetic pole at 28°S/161.7°W. The positions of these poles on the apparent polar wander path (APW) for Africa indicate that the critical zone had acquired its remanent magnetization before the main zone. Fold tests prove that the main zone in the eastern Bushveld Complex had acquired its remanent magnetization with the igneous layering in a horizontal position.

INTRODUCTION

Gough and Van Niekerk (1959) presented palaeomagnetic data from five sampling sites in the main zone of the Bushveld Complex. One of these sites were situated in the eastern Bushveld Complex near Roossenekal. From this site, 23 core samples were obtained, which yielded a mean palaeomagnetic magnetization direction with a declination (D) of 36.8° and an inclination (I) of 55.3° (Gough and Van Niekerk, 1959, p. 131). The rock unit which was sampled is now formally known as the Leolo Mountain Gabbro-Norite (S.A.C.S., 1980) or informally, as subzone B of the main zone (Fig. 1).

The sampling pattern of the present study yielded twenty sampling sites covering subzones A, B and C of the main zone in the eastern Bushveld Complex (Fig. 1). Gabbro, norite and anorthosite, exhibiting very little, if any, alteration, are the main rock types. The relative freshness of the samples is due to siting all sampling sites in quarries or road cuttings.

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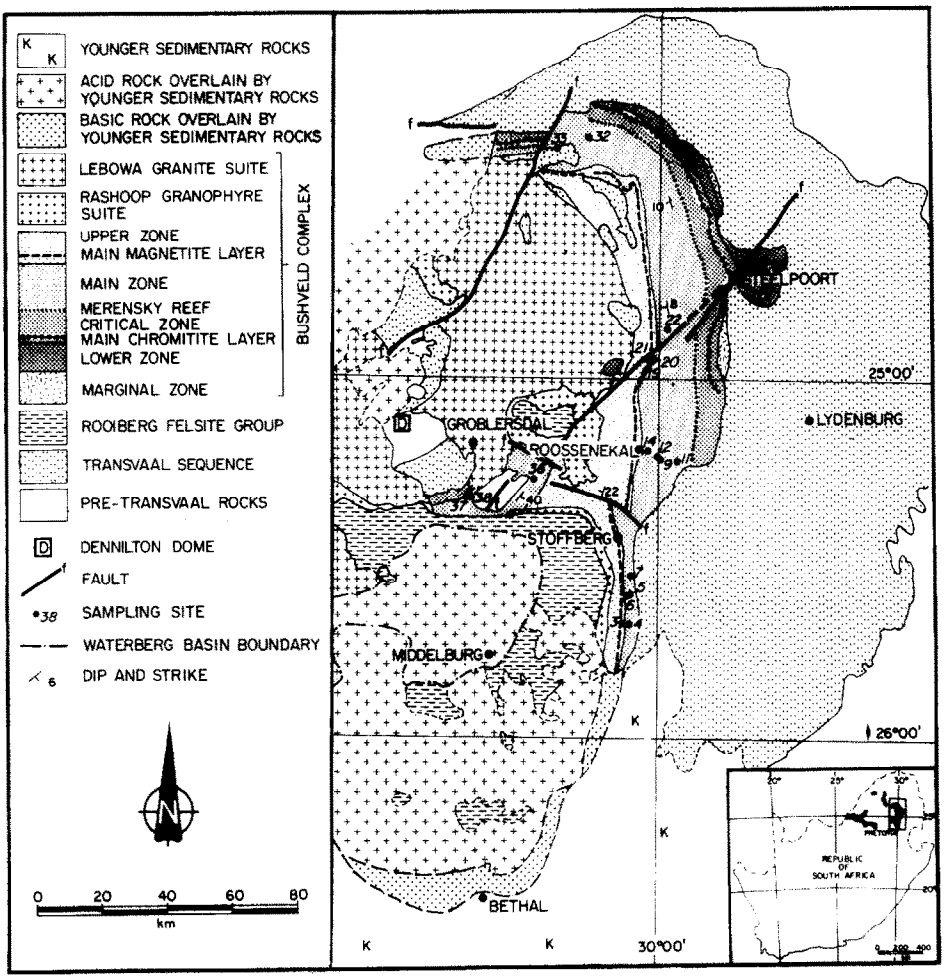


Fig. 1. Geological map of the eastern Bushveld Complex showing sampling sites in the main zone.

Samples were taken with a petrol driven portable core drill. The core samples are 25 mm in diameter and up to 250 mm in length. Four to five core samples were taken at each sampling site. Each core was cut into an average of five cylindrical specimens. Core orientation was done with a sun compass (Creer and Sanver, 1967) which allows orientation to a precision of 1° (Gough, 1967).

LABORATORY MEASUREMENTS

The remanent magnetization of specimens was measured using a Digico "Complete Results" fluxgate spinner magnetometer. In practice it was found that to measure a reliable direction of magnetization to within one degree, a minimum intensity of magnetization of 0.5×10^{-3} A m^{-1} was required.

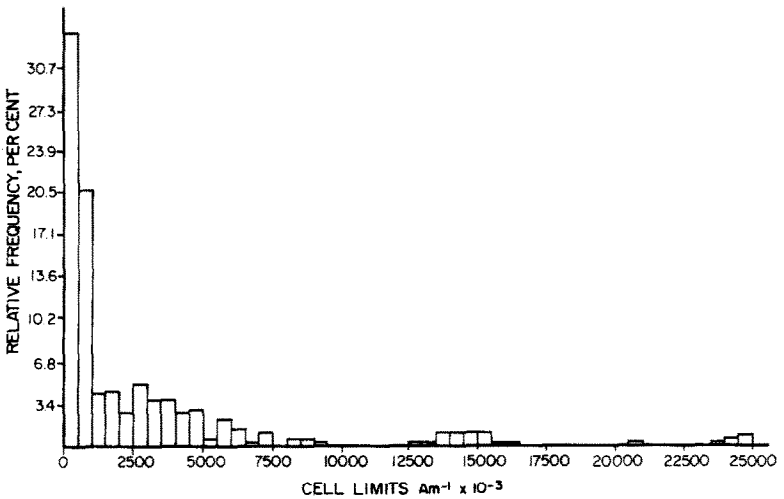


Fig. 2. Histogram plot of the NRM intensities of specimens from the main zone. Cell width is 500×10^{-3} A m^{-1}.

TABLE 1

Natural remanent magnetization directions of sites in the main zone, eastern Bushveld Complex *

Site	N	D (°)	I (°)	α_{95} (°)	k
3	4	34.5	58.1	1.9	2330
4	4	29.9	59.6	4.3	461
5	3	30.0	60.0	6.5	351
6	4	29.5	59.7	5.5	281
7	3	28.5	65.2	11.3	120
9	4	33.2	60.7	8.2	127
12	4	30.6	59.7	4.0	537
19	4	195.0	−58.8	21.5	34
20	4	16.7	−41.2	8.9	195
21	4	353.7	−14.0	7.0	171
22	4	220.0	−80.0	9.7	90
32	4	21.0	43.0	4.4	432
36	4	354.0	61.9	4.2	496
37	4	8.7	39.5	6.9	177
38	4	358.7	41.7	8.6	114

* N is the number of core samples; D is the declination of the magnetization direction, measured eastwards from north; I is the inclination of the magnetization direction, downwards positive, upwards negative; α_{95} is the semi-angle of cone of 95% confidence ; k is the estimate of the precision parameter κ.

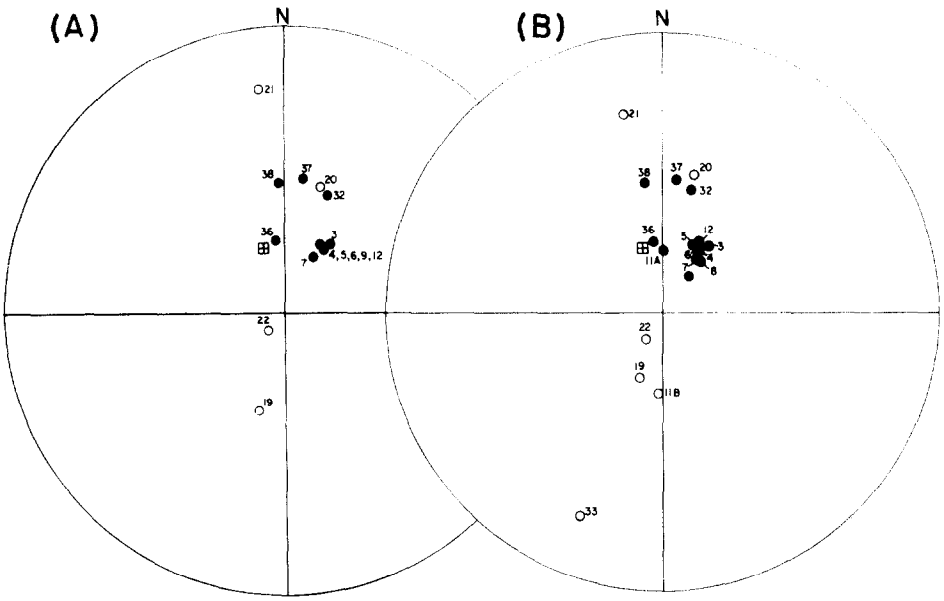


Fig. 3. A. Equal angle projections of NRM directions of sites. B. Magnetization directions of sites, after AF demagnetization. ● = lower hemisphere, ○ = upper hemisphere. Present-day magnetic field direction upward indicated by ⊞.

Continuous thermal demagnetization of specimens was done with a slightly-modified form of the spinner magnetometer. It was possible to measure remanence at high temperatures without the need for cooling the specimen in a field free space. Operating the instrument in this mode, however, led to a decreased sensitivity, and it was found that the minimum intensity of magnetization that could be measured reliably was found to be in the region of 4×10^{-3} A m^{-1}.

Alternating field (AF) demagnetization of specimens was done with an apparatus built at the Geological Survey of South Africa following a description by McElhinny (1964a). This apparatus is capable of creating a peak alternating magnetic field of 100 mT.

The mean intensity of the NRM of all specimens from the main zone in the eastern Bushveld Complex is 3395×10^{-3} A m^{-1}. A histogram plot of the intensities of these specimens is presented in Fig. 2.

Five sites had random NRM directions at sample level according to the criterion of Watson (1956) at 95% confidence limit. The NRM data of the remaining sites are listed in Table 1 and the NRM directions of these sites are shown in Fig. 3A.

If all sites with positive inclinations of magnetization directions are grouped together, group BMZ1 forms with the following group statistics: $N = 11$; $D = 20.4°$; $I = 56.2°$; $\alpha_{95} = 7°$; $k = 44$ with N the number of sites, D and I the declination and inclination of the group mean magnetization direction, α_{95} the radius of the circle of

95% confidence around the mean direction, and k, the estimate of the precision parameter κ.

Alternating field demagnetization

Bulk AF demagnetization of specimens was done at optimum AF demagnetization values indicated by the Briden Stability Index (Briden, 1972) for each site. The results are listed in Table 2 and the magnetization directions are shown in Fig. 3B.

Three sites 11, 14 and 33, which had inconsistent NRM directions at sample level, grouped significantly after AF demagnetization of specimens. The directions from site 11 are bipolar and the site has correspondingly been split into two sub-sites, 11A and 11B.

TABLE 2

Magnetization directions of sites in the main zone, eastern Bushveld Complex after bulk alternating field demagnetization (A), and (B) with the igneous layering in a horizontal position *

Site	Alternating field (mT)	A				B	
		D (°)	I (°)	α_{95} (°)	k	D ** (°)	I (°)
3	10	36.2	57.6	2.2	1775	23.6	60.0
4	20	29.6	59.3	4.4	444	15.9	61.1
5**	20	27.5	58.2	–	–	11.0	60.7
6	10	26.4	60.6	6.2	219	4.2	63.0
7**	20	32.6	60.8	–	–	18.6	64.3
9	20	35.7	64.4	8.7	111	9.3	67.5
11A **	40	1.4	65.0	–	–	339.2	63.7
11B **	40	182.4	−58.4	–	–	172.6	−58.2
12	60	38.4	64.2	4.8	695	12.3	67.8
14	60	35.8	71.5	19.4	23	0.3	75.0
19	20	198.6	−62.9	17.0	53	156.4	−54.0
20	40	13.8	−37.2	7.6	258	32.6	−37.9
21	20	348.7	−19.3	11.0	70	41.8	−16.5
22	20	211.5	−77.5	9.9	86	155.0	−59.7
32	10	13.8	41.9	8.4	120	4.6	48.2
33	20	202.7	−14.7	26.0	23	191.2	−48.7
36	0	354.0	61.9	4.2	496	29.0	67.5
37	10	6.7	39.5	7.8	140	19.9	68.2
38	20	352.5	40.8	10.5	76	349.3	70.0

* N is the number of core samples; D is the declination of the magnetization direction; measured eastwards from north; I is the inclination of the magnetization direction; downwards positive, upwards negative; α_{95} is the semi-angle of cone of 95% confidence; k is the estimate of the precision parameter κ.
** $N = 2$.

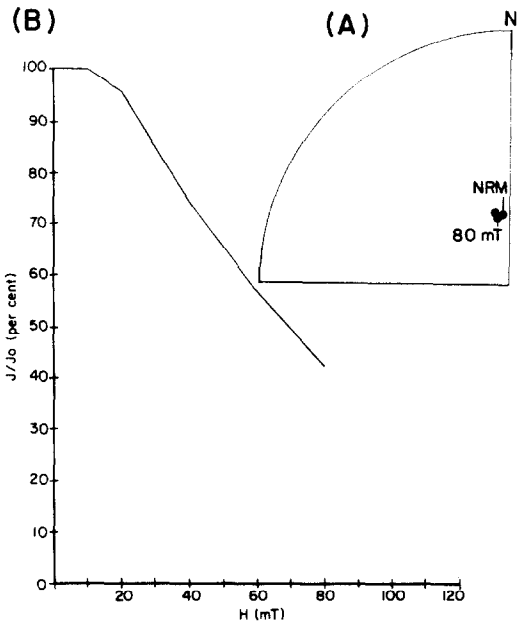


Fig. 4. A. Stepwise AF demagnetization results of specimen 36/4C. Maximum AF value 80 mT. Plotting convention as in Fig. 3. B. Normalized intensity response curve of specimen 36/4C during AF demagnetization. This curve illustrates the "hardness" of magnetization which is typical of most specimens from subzone B.

Thirteen sites are reversely magnetized (positive inclinations). These sites form group BMZ2AF with $N = 13$; $D = 18.8°$; $I = 58.5°$; $\alpha_{95} = 7.0°$; $k = 36$. The mean magnetization direction of this group does not differ much from that of group BMZ1. Apart from two new sites (11A and 14) included in BMZ2AF, it is evident that the bulk AF demagnetization failed to improve the grouping of the majority of the magnetization directions from the different sites. The stepwise demagnetization of specimens from this group produced very little or no change in the magnetization directions with the behaviour of specimen 36/4C, shown in Fig. 4A as a typical example.

The intensity response curve of the same specimen (Fig. 4B) during stepwise AF demagnetization, revealed a stable and very "hard" magnetization, with more than 40% of the magnetization remaining after demagnetization in a field of 80 mT. This type of "hard" stable magnetization has been found by other authors e.g. in the Modipe Gabbro (Evans et al., 1968) and in the Lambertville diabase (Hargraves and Young, 1969). The source of this stable "hard" magnetization was in both cases shown to be single domain magnetite grains, showing extensive shape anisotropy.

It is interesting to note further that all the sites in group BMZ2AF, with the exception of site 11A, are situated in subzone B of the main zone, i.e. the Leolo

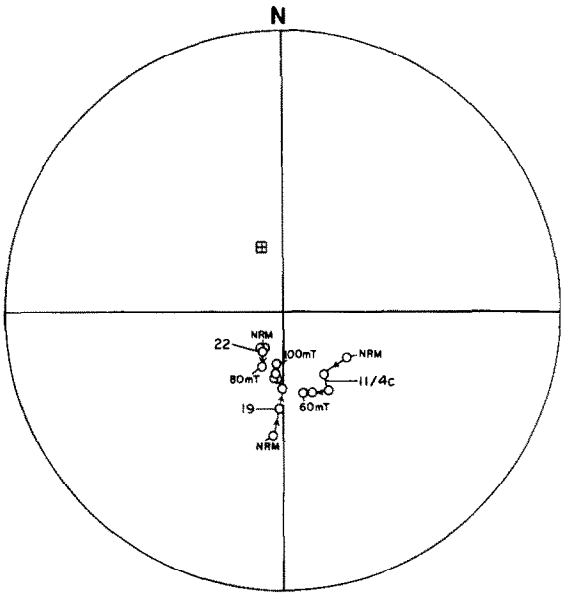


Fig. 5. Stepwise AF demagnetization of specimens from sites 11B, 19 and 22. Plotting convention as in Fig. 3.

Mountain Gabbro-Norite. Site 11 is situated at the base of the main zone within subzone A i.e. the Winnaarshoek Norite-Anorthosite.

At three sites (11B, 19 and 22) where the magnetization directions are normal (negative inclinations) the magnetization directions converge during step-wise AF demagnetization (Fig. 5) and a fourth site (33) shows a significant sample grouping of magnetization directions (negative inclination) after bulk AF demagnetization. These four sites form group BMZ3AF, with $N = 4$; $D = 197.8°$; $I = -54.3°$; $\alpha_{95} = 32.8°$; $k = 9$.

With the radius of the circle of 95% confidence of this group being 32.8° and an estimate of the precision of 9, the grouping is poor, but this may be due to the effect of the attitude variations of the layered sequence at the sites.

Two other sites, 20 and 21, also show magnetization directions with negative inclinations. These directions changed very little during stepwise AF demagnetization, and after bulk AF demagnetization failed to group with the sites in group BMZ3AF. The reason for this is not apparent, especially because with the exception of site 11B, all the sites normally magnetized are situated in subzone C of the main zone.

The magnetization directions obtained with bulk AF demagnetization and then corrected for the dip at each sampling site, are listed in Table 2.

These directions are plotted in Fig. 6 and group BMZ2AFR emerges with the

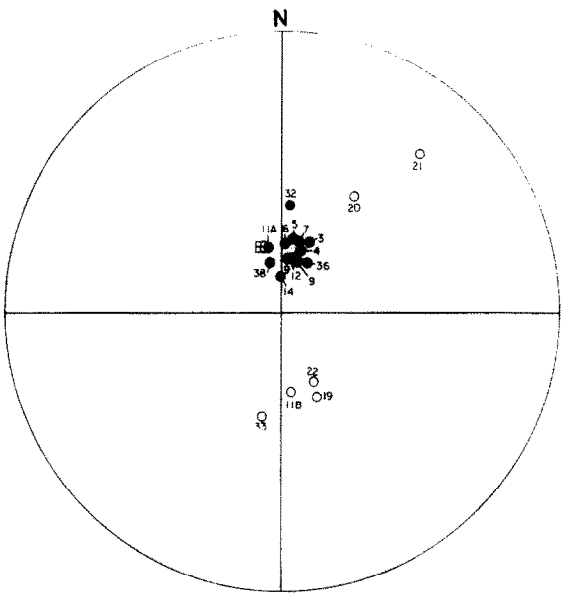


Fig. 6. Magnetization directions of sites after AF demagnetization, with the igneous layering in a horizontal position. Plotting convention as in Fig. 3.

following statistics: $N = 13$; $D = 10°$; $I = 64.7°$; $\alpha_{95} = 4.2°$; $k = 100$.

The equivalent of group BMZ2AFR, uncorrected for dip, is group BMZ2AF, and from a comparison of the group statistics it is clear that the fold test has resulted in a much better grouping of magnetization directions in group BMZ2AFR. The increase in the value of k from 36 to 100 is significant at even the 99% confidence limit according to the test of McElhinny (1946b).

The circular standard deviation of group BMZ2AFR is 8.1° with upper and lower limits of 11.3° and 6.3° at the 95% confidence limit (Cox, 1969), against an expected value of 11.7° for a palaeolatitude of 46.6° (Brock, 1971). According to Opdyke (1972) 27,000 years are needed in relative recent geological times to average out palaeosecular variation. If one assumes a similar time span for the Precambrian, it follows that the Curie isotherm must have moved through that part of the main zone represented by the sites in group BMZ2AFR over a period of less than 27,000 years.

After rotation, group BMZ3AF has the following statistics: $N = 4$; $D = 169.8°$; $I = -56.0°$; $\alpha_{95} = 12.6°$; $k = 54$.

The improvement in k from 9 to 54 is significant at the 95% confidence limit (McElhinny, 1964b). There can be little doubt now, that the main zone of the layered sequence in the eastern Bushveld Complex, acquired its present dip and dip direction only after cooling, to a temperature below its Curie point.

Thermal demagnetization

The stable behaviour exhibited by specimens belonging to group BMZ2AF during AF demagnetization was manifested by different specimens from the same group during continuous thermal demagnetization (Fig. 7).

The intensity response curves of two specimens (Fig. 8), which can be regarded as typical of all these specimens display distributed blocking temperature spectra but show a well-defined Curie point at 575°C. This Curie point indicates magnetite as the primary source of remanent magnetization in these specimens.

The magnetization direction of two sites (11A and 14), from group BMZ2AF, which became consistent with directions from other sites in this group only after AF demagnetization, as well as the magnetization directions from site 13, which up to this stage of the investigation has been rejected due to inconsistency of sample directions, converged during continuous thermal demagnetization towards the mean direction of group BMZ2AF (Fig. 9). This is additional evidence that, although uncorrected for dip of the layered sequence, the mean magnetization direction of group BMZ2AF represents a primary magnetization direction.

The intensity of magnetization of specimens from sites 11B and 33 are too low to allow thermal demagnetization of the specimens with the present equipment. In Fig. 9, however, the directional response to thermal demagnetization of specimens from

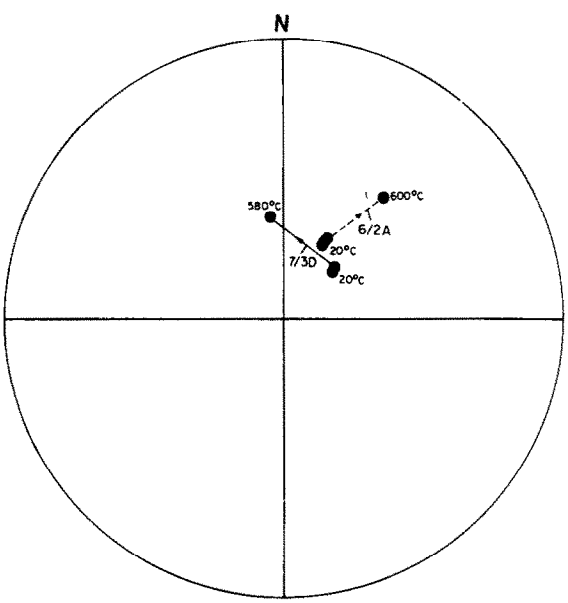


Fig. 7. The change in magnetization directions of two specimens from BMZ2AF, subzone B, main zone, during continuous thermal demagnetization. Plotting convention as in Fig. 3.

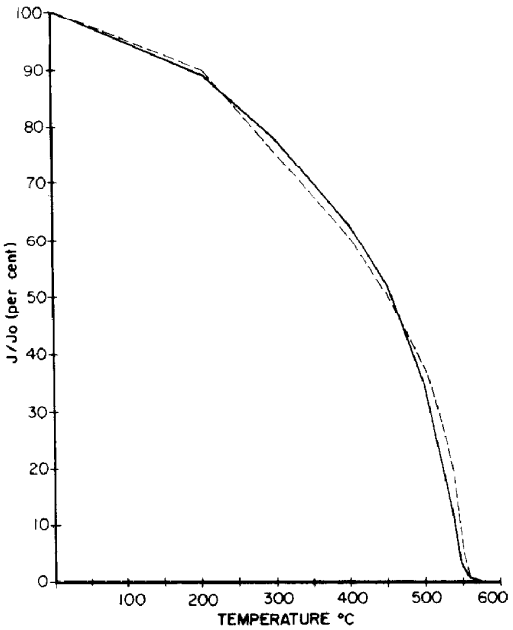


Fig. 8. Typical normalized intensity response curves during continuous thermal demagnetization of two specimens from group BMZ2AF which are situated in subzone B, main zone.

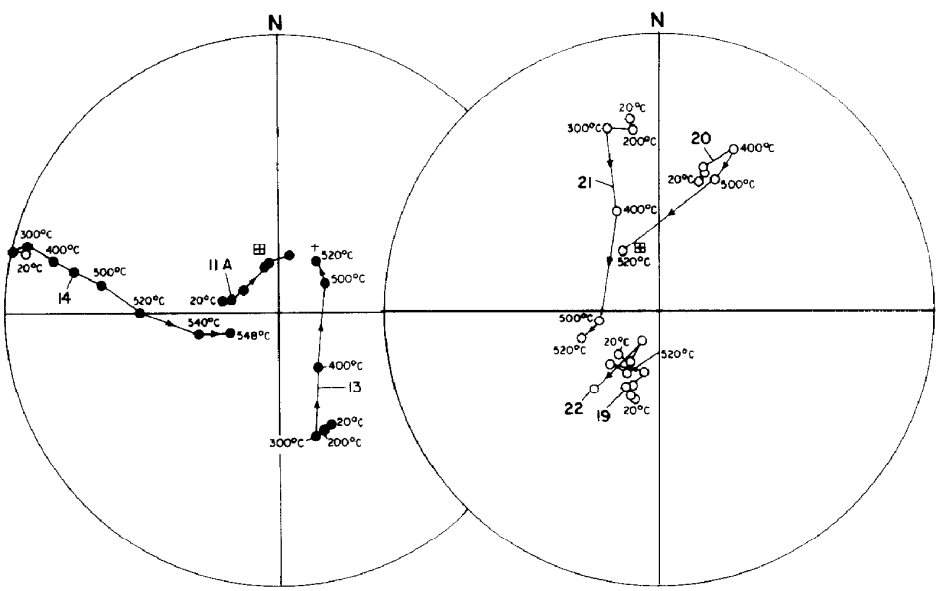


Fig. 9. The change in magnetization directions of specimens from sites 11A, 13, 14, 19, 20, 21 and 22, during continuous thermal demagnetization. Plotting convention as in Fig. 3.

sites 19, 20, 21 and 22 are shown. The remanent remagnetization direction of sites 19 and 22, which grouped well with group BMZ3AF, display a tendency to move closer together during thermal demagnetization (Fig. 9). The movement is not much and does not follow circles of remagnetization. From this it would seem that the thermal demagnetization succeeds in removing numerous very small viscous remanent magnetizations (VRM's) resulting in a closer grouping of magnetization directions at high temperatures.

The remanent magnetization directions of two sites (20 and 21), which proved to be stable during AF demagnetization, but incompatible with either group BMZ2AF or group BMZ3AF, change rapidly at high temperatures (Fig. 9). Although the change of the vector directions does not describe circles of remagnetization, it is clear that the directions move towards the mean direction of magnetization of group BMZ3AF. This behaviour during demagnetization (thermal and AF) can only be explained by the presence of magnetization components with high coercivities and low blocking temperatures, which mask the primary magnetization direction at low temperatures and during AF demagnetization.

Sites 19, 20, 21, 22 and 33 are situated in subzone C of the mean zone. The mean magnetization direction is obtained by grouping directions from sites 19, 22 and 33 together (obtained by AF demagnetization) which represents the mean magnetiza-

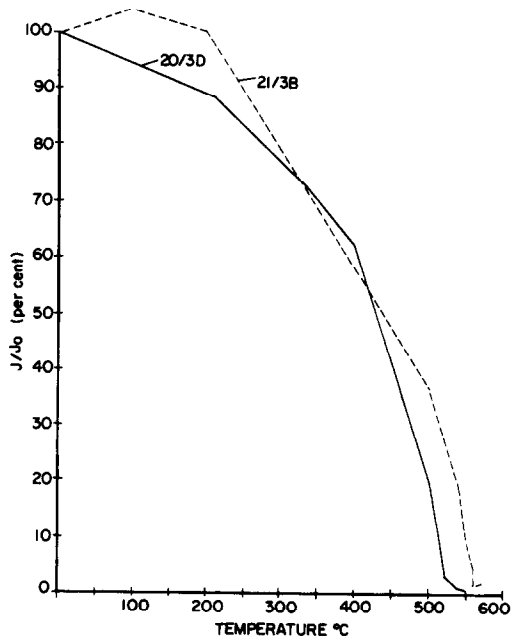


Fig. 10. Normalized intensity response curves of two specimens from sites 20 and 21 during continuous thermal demagnetization.

tion direction of subzone C, or the Mapoch Gabbro-Norite as it is formally known. Uncorrected for dip, this group of three sites yields the following rather poor, statistical data. Group: BMZ4AF: $N = 3$; $D = 202.7°$; $I = -52.3°$; $\alpha_{95} = 54°$; $k = 6$.

However, with the layering restored to horizontal, the statistics improve considerably to: $N = 3$; $D = 168.9°$; $I = -55.0°$; $\alpha_{95} = 20°$; $k = 37$.

For this last group the fold test (Graham, 1949) is significant at the 90% confidence limit. It is further clear that these two groups can only be accepted on the basis of the additional evidence supplied by the thermal demagnetization of specimens from sites 20 and 21.

The thermal intensity response curves of two specimens, 20/3D and 21/3B (Fig. 10) are fairly typical of a multicomponent remanent magnetization. The curve of specimen 20/3D suggests the existence of two Curie points, one at 525° and the other at 550°C. This feature is, however, absent on the curve of specimen 21/3B, which indicates a Curie point at approximately 570°C.

MINERALOGY OF OPAQUE MINERALS

Specimens from subzone B of the main zone are characterized by the presence of magnetite, ilmenite and various sulphide minerals. A preliminary investigation of polished thin sections with an ore microscope, however, failed to identify any opaque grains as magnetite. If magnetite were not present, it would be very difficult to explain the observed Curie point of 575°C of these specimens. This combined with the previously described "hard" (Fig. 4) magnetization, which in other studies was found to be due to the existence of fine magnetite needles in either pyroxenes (Evans et al., 1968) or in plagioclase grains (Hargraves and Young, 1969), led to a more detailed examination of the pyroxene and plagioclase grains under maximum magnification with reflected and transmitted light microscopy. The result of this was, that opaque, needle-like grains were indeed found in both the pyroxene and plagioclase crystals (Figs. 11 and 12). The opaque grains in the pyroxene crystals are typically needle-like with an average width of approximately 1.5 μm, and a length to width ratio of 3 : 1, but occasionally larger lamellae are present. What appear to be long thin needles in some cases, can also be thin plates being looked at edge-on. The opaque grains observed in the plagioclase crystals are very similar, apart from being a little smaller with an average width of approximately 1 μm, and more elongated with a length to width ratio of up to 10 : 1.

The identification of these lamellae proved to be very difficult because of their small size, and only some of the larger opaque grains in the pyroxene crystals enabled their positive identification as magnetite with the aid of a scanning electron microscope. The presence of magnetite as needles in plagioclase crystals of the gabbros from the main zone of the layered sequence in the eastern Bushveld Complex was amongst others recognized by Groeneveld (1970) and Scharlau (1972),

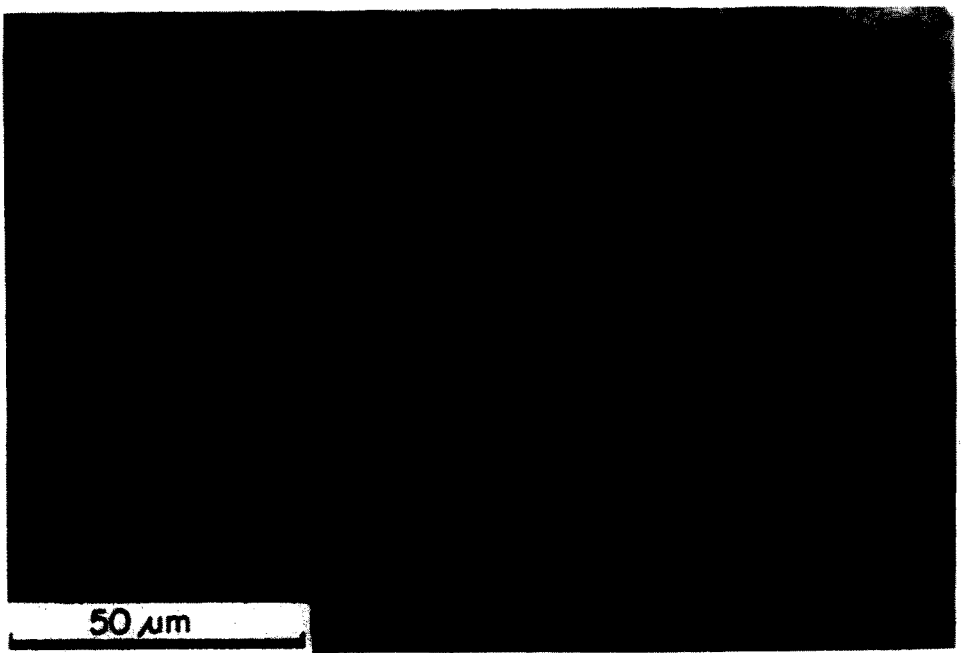


Fig. 11. Magnetite lamellae in a pyroxene crystal, subzone B, main zone. Transmitted light, magnification 776×.

although earlier workers argued that these opaque grains could possibly also be needles of rutile and ilmenite (Brandt, 1946; Willemse and Viljoen, 1970). It was further observed, that the more abundant these needles appeared in a thin section, the higher the intensity of remanent magnetization of the specimen. Dunlop (1981), pointed out that this type of very "hard" remanent magnetization can only be explained by the presence of single domain grains (grain size = 1 μm; Soffel, 1971), which are characterized by high degree of shape anisotropy. From this evidence it is concluded that the main carrier of remanent magnetization in these specimens, is the observed magnetite lamellae in the pyroxene and plagioclase crystals.

The genesis of magnetite needles in pyroxene crystals from the Grenville province is described by Fleet et al. (1980), who argue that the needles exsolved from the parent pyroxene crystal at high temperatures.

Similarly, for the opaque grains in the plagioclase crystals from the Bushveld Complex, Scharlau (1972) described an exsolution process at high temperatures which took place shortly after crystallization of the plagioclase from the magma. This indicates that the remanent magnetization of these exsolution lamellae is a primary magnetization, associated with the crystallization and subsequent cooling of subzone B of the main zone.

Ilmenite occurs as discrete grains up to 0.5 mm in diameter. The ilmenite is



Fig. 12. Magnetite lamellae in a plagioclase crystal subzone B, main zone. Transmitted light, magnification 776×.

generally unaltered, and the presence of haematite exsolution lamellae in ilmenite is very rare. The total ilmenite content is estimated to be less than 1 vol.%. Electron microprobe analyses of ilmenite grains indicated a composition of approximately 85% $FeTiO_3$ and 15% Fe_2O_3. In the composition range 45–95% ilmenite the haemo-ilmenites are ferromagnetic (O'Reilly, 1984) but with Curie temperatures generally below room temperature.

Several types of sulphides are present in the specimens, some of which have been identified as pyrrhotite. These grains are smaller and less abundant than the ilmenite grains. Pyrrhotite is a ferromagnetic mineral with a Curie point between 300°C and 325°C (Nagata, 1953), and will contribute to the remanent magnetization of specimens.

The three sites (19, 22 and 33) in group BMZ4AFR as well as sites 20 and 21, which are compatible with this group on the basis of results from the thermal demagnetization, are characterized by a very similar opaque mineralogy and to a certain extent the same 'hard" NRM (Fig. 13) as sites in group BMZ2AFR. Magnetite exsolution lamellae are present in both pyroxene and plagioclase crystals although less abundant, whereas discrete grains of magnetite are very rare. Ilmenite, as well as pyrrhotite are present as smaller grains. In the case of subzone C, however, more ilmenite grains with hematite exsolution lamellae are present.

Although the magnetization direction of site 11B can be grouped with group

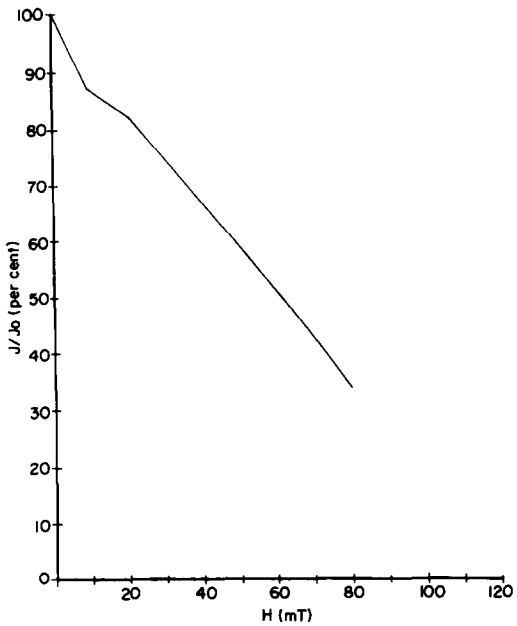


Fig. 13. Normalized intensity response curves of specimen 20/1C of subzone C, main zone, during AF demagnetization.

BMZ4AF, it has a marked different opaque mineralogy to both the sites in groups BMZ4AF and BMZ2AF including site 11A. The specimens are characterized by a lack of opaque minerals and neither tiny magnetite rods nor discrete magnetite grains could be found in the polished thin sections with an ore microscope. The absence of magnetite correlates well with the observed low NRM intensity. Pyrrhotite grains are present in small quantities; however, more striking is the presence of chromite grains.

Magnetite was never identified optically in the polished thin sections, but in search of chromite grains, opaque grains were found which contain no Cr but a fairly high percentage of Ti, which indicates that the grains are either magnetite or ilmenite.

The presence of chromite is further a possible indicator that the geological location of this site (subzone A of the main zone) has been wrongly identified, and that the site could be situated in the critical zone. Both Prof. G. von Gruenewaldt and M.R. Sharpe (pers. commun., 1983) confirmed the site position as being near the base of the main zone and definitely not in the critical zone.

MIXED POLARITY OF SITE 11

Specimens from site 11 (sub-sites 11A and 11B) exhibited a complex change in magnetization directions during stepwise AF and continuous thermal demagnetiza-

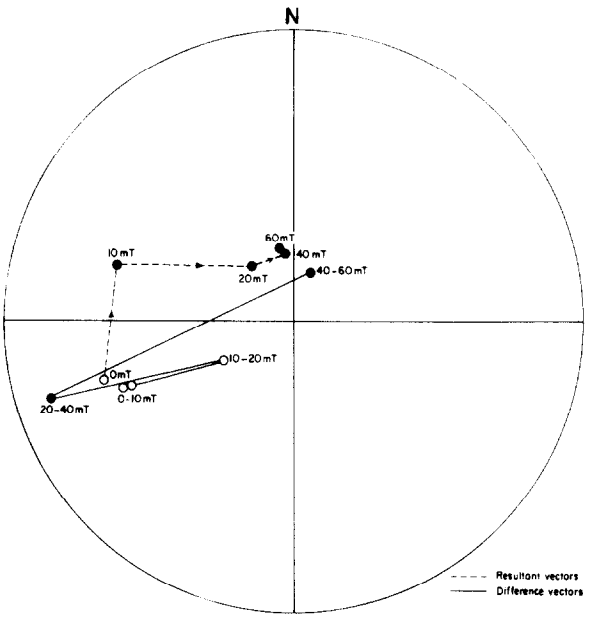


Fig. 14. AF demagnetization response of specimen 11/3B (site 11A). Resultant and difference vectors are shown. Difference vectors represent magnetization vector destroyed between each AF step. Plotting convention as in Fig. 3.

tion. With the aid of the Briden SI (Briden, 1972) alternating demagnetization fields were found for the bulk AF demagnetization of specimens from the site. The bulk AF demagnetization successfully located two magnetization directions. The first direction in core samples 11/1 and 11/3 is in agreement with the mean magnetization direction of group BMZ2AF, and is now referred to as the A magnetization direction. The second magnetization direction, which can be called the B direction, was isolated in core samples 11/4 and 11/5, and this magnetization direction is in agreement with the mean direction of group BMZ4AF. For grouping purposes site 11 was then split into two sites, 11A and 11B, representing the A (reversed) and B (normal) field magnetization directions respectively.

In an attempt to decide whether the approximately antipodal A and B magnetization directions represent a geomagnetic field-reversal or a self-reversal of magnetization, extensive analyzes of thermal and AF demagnetization data, as well as various experiments were undertaken, the results of which will be presented and analysed in the following paragraphs.

The change in magnetization directions of specimen 11/3B during stepwise AF demagnetization and the magnetizations removed after each step, are shown in Fig. 14. In the initial AF demagnetization steps, a magnetization component is removed with a direction approximately parallel to the NRM directions of the specimen (say

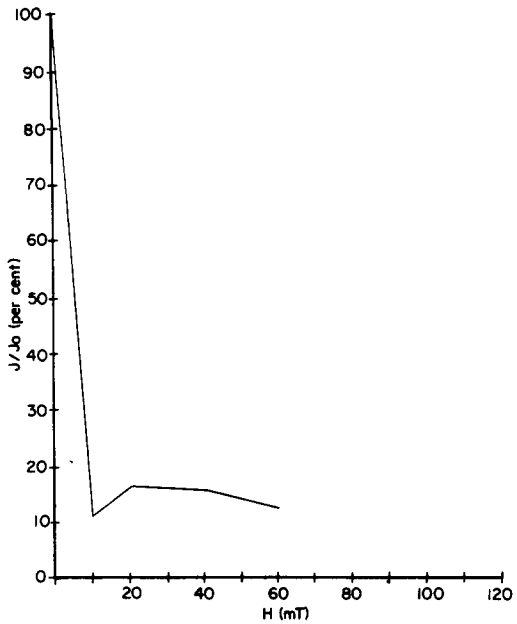


Fig. 15. Normalized intensity response curve of specimen 11/3B (site 11A) during AF demagnetization.

the C direction). This, however, changes rapidly as the AF values increase, and at the 40 mT to 60 mT step a direction broadly corresponding to the A direction is removed. At this stage the resultant and the difference vectors converge, indicating that only one component is removed. The AF demagnetization intensity curve of this specimen is shown in Fig. 15, from which it is clear that two more or less opposing vectors are removed during AF cleaning. The first vector is destroyed at low AF values, manifested by a sharp decrease in the intensity of magnetization, but from 10 mT the intensity actually increases, reaches a maximum at 20 mT, from where, at higher AF values, the second component is destroyed. These data clearly indicate that two magnetization directions C and A reside in the specimens from site 11A, the C direction being the lower coercivity component.

The directional response of specimen 11/4C (site 11B) with the difference vector removed during stepwise AF demagnetization is shown in Fig. 16. Again, at low AF values a magnetization direction is removed, which broadly corresponds to the NRM direction of the specimen and which is similar to the C direction removed in specimen 11/3B. At higher AF values the resultant vector changes rapidly, whereafter it follows a circle of remagnetization converging to the B magnetization direction. The difference vector indicates that at intermediate AF values a direction corresponding to the A magnetization direction is removed, and only at 60 mT do the difference and resultant vectors converge, indicating the removal of the B magnetiza-

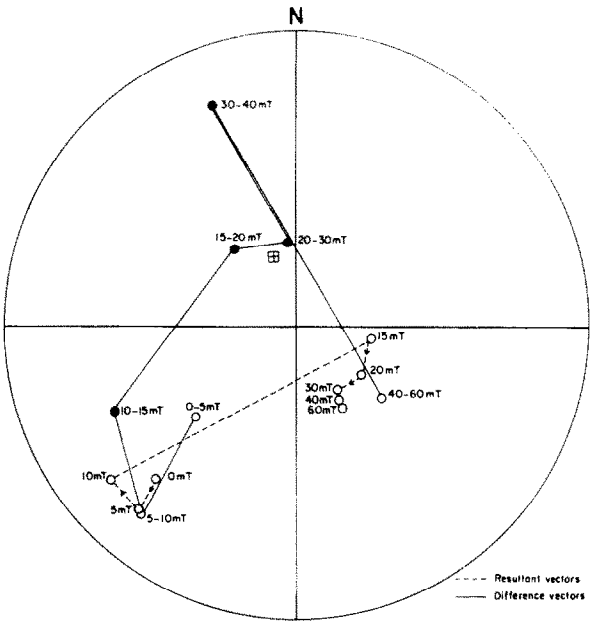


Fig. 16. AF demagnetization response of specimen 11/4C (site 11B). Resultant and difference vectors shown. Plotting convention as in Fig. 3.

tion direction. Present in this specimen there are three magnetization components, C, A and B, in order of their relative stability.

The C direction of magnetization can be regarded as either an IRM or a VRM, because it is easily removed at relatively low AF values, but the A and B directions must, at this stage, be regarded as primary magnetization components of sites 11A and 11B respectively. This indicates then that the bipolarity of site 11 is due to primary magnetization directions.

The directional response of specimen 11/3C to thermal demagnetization is shown in Fig. 17. The difference vector here again indicates the presence of two magnetization directions C and A, similar to that indicated by the AF demagnetization. The C direction is a low temperature component, while the A direction emerges just before the Curie point of the specimen is reached. The same data for specimen 11/1B (Fig. 17) again show two magnetization components, an A direction, and a direction corresponding to the C direction, but with the inclination changed from negative to positive. Either this illustrates the capability of specimens from sample 11/1 to undergo a partial self-reversal of magnetization, or it is possibly due to an IRM, caused by lightning which affected samples 11/1 and 11/3 differently.

The intensity of the NRM of specimens from site 11B (samples 11/4 and 11/5) is too low to allow thermal demagnetization of the specimens, and consequently no further information on the blocking temperature spectra is available.

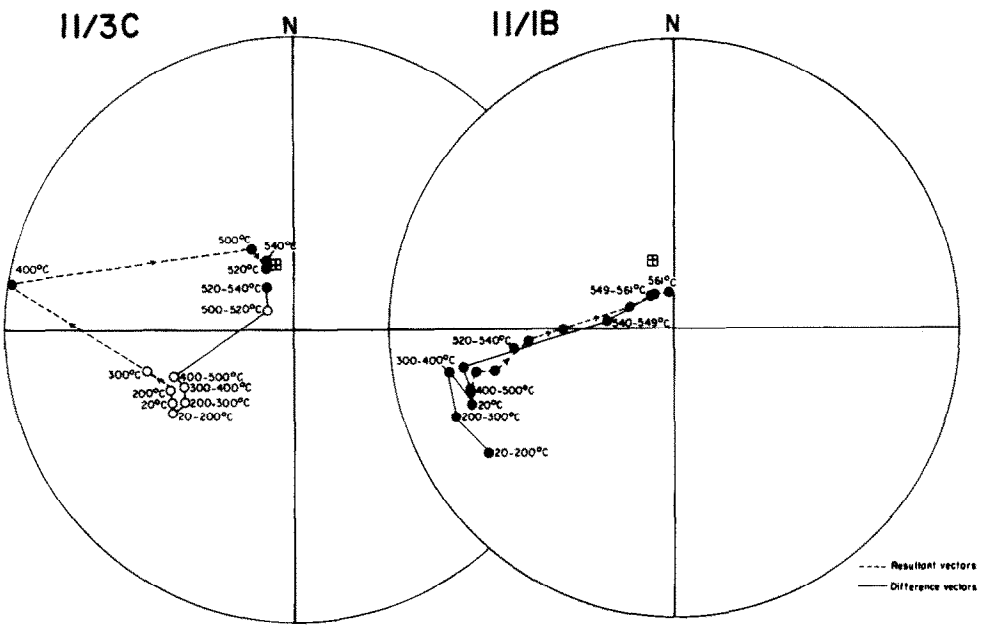


Fig. 17. Thermal demagnetization of specimens 11/3C and 11/1B. Both specimens are from site 11A. Resultant and difference vectors shown. Plotting convention as in Fig. 3.

One specimen from each sample at site 11 was heated in the presence of nitrogen to 650°C and allowed to cool in the earth's magnetic field. All the specimens, with the exception of 11/5E, acquired a magnetization parallel to the magnetic field in which they cooled. Specimen 11/5E acquired a magnetization direction of which the declination differed by 90° to the field in which it cooled, but the inclination was in the same direction as that of the magnetic field in which it cooled.

The AF demagnetization response of the artificial TRM given to this specimen is shown in Figs. 18 and 19. The change in vector direction during stepwise AF demagnetization of the specimen, shows that the TRM was acquired in such a way that it represents a multicomponent magnetization. The intensity response curve (Fig. 19) has a shape broadly similar to the intensity response curves obtained by AF demagnetization of the NRM of specimens from site 11B (Fig. 19).

Another specimen from site 11A with an artificial TRM, was thermally demagnetized. Because this specimen had acquired a magnetization parallel to the field in which it cooled down it was thought that the magnetization direction should change very little during thermal demagnetization. This was not the case, and during continuous thermal demagnetization the magnetization direction exhibited a swinging behaviour (Fig. 20). The reason for this observed behaviour is not clear, but it is possibly an indication of the ability of specimens from site 11, to acquire magnetization components which are not parallel to the field in which they acquire a TRM.

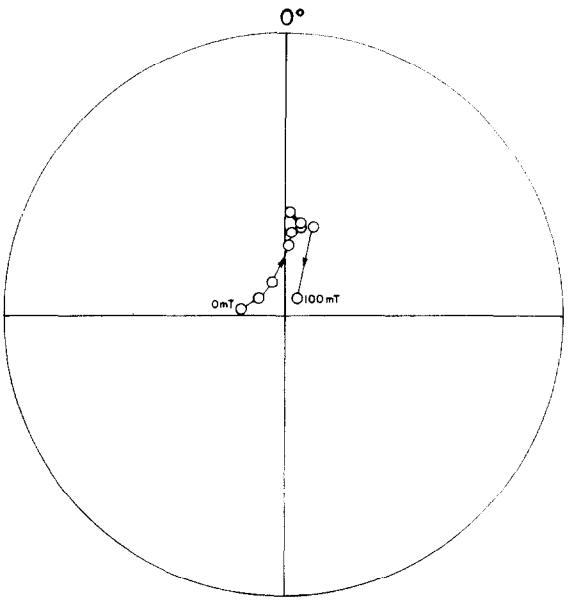


Fig. 18. AF demagnetization of specimen 11/5E (site 11B), after the specimen had been given an artificial TRM in the earth's magnetic field. Plotting convention as in Fig. 3.

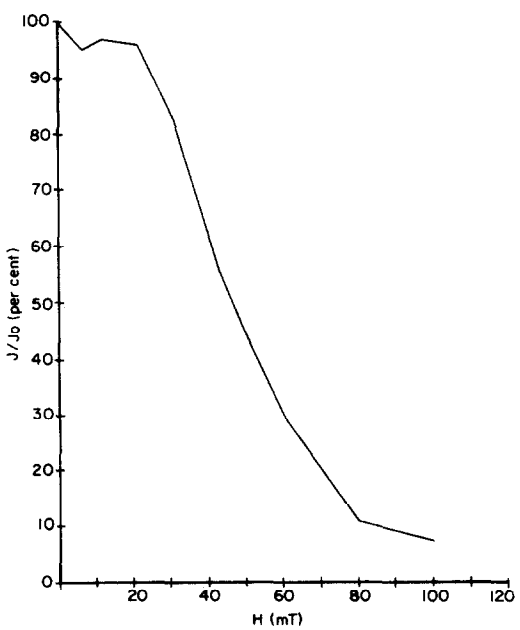


Fig. 19. Normalized intensity response curve of specimen 11/5E (site 11B) during AF demagnetization, after the specimen had been given an artificial TRM in the earth's magnetic field.

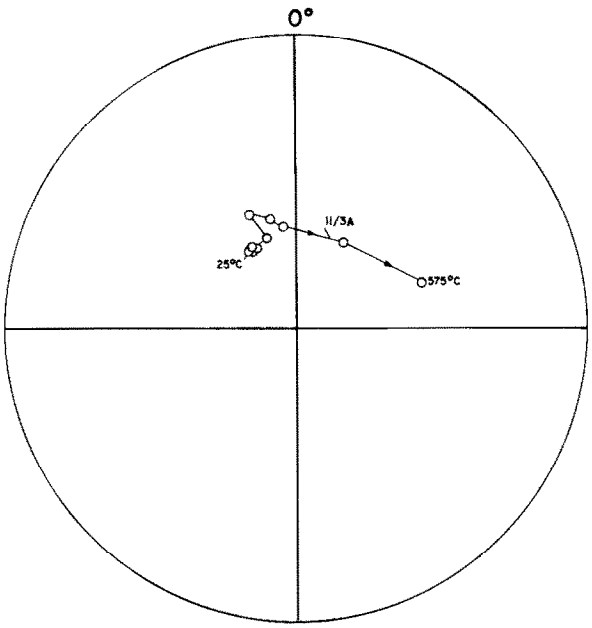


Fig. 20. Thermal demagnetization of specimen 11/3A (site 11A), after the specimen had been given an artificial TRM in the earth's magnetic field. Plotting convention as in Fig. 3.

To test the stability of magnetization of site 11 further, the measurement of the NRM of a number of specimens was repeated after 12 months. All the magnetization directions remained the same, except for one specimen from site 11B. In this case the inclination of the magnetization relative to a fiducial mark on the specimen, changed from $-45°$ to $+45°$. Although it is possible that the specimen had acquired a VRM in the laboratory during the twelve months, no indication of it could be found, because the intensity of magnetization remained exactly the same.

Despite the previous slight evidence in favour of a self-reversal of magnetization, the fact that all the measured specimens from site 11B also contain the A magnetization direction of site 11A, can be used to argue in favour of a geomagnetic field-reversal. If a field reversal had taken place immediately after the Curie isotherm has passed through site 11B, it is conceivable that the samples from site 11B, at that stage at a relatively high temperature, could have acquired at least a component of magnetization parallel to the new (reversed) ambient magnetic field. A similar argument to explain mixed polarities at one site in the Limpopo belt, is used by Morgan and Briden (1981) in support of a geomagnetic field-reversal. However, these authors give no details on the mineralogy of the specimens, which in the case of sites 11A and 11B do differ.

POLARITY DIFFERENCE OF MAGNETIZATION BETWEEN SUBZONES B AND C

The mean magnetization direction of subzone B of the main zone in the eastern Bushveld Complex is reversed as given by group BMZ2AFR, and that of subzone C is normal as given by group BMZ4AFR. These different polarities can be regarded as approximately antipodal. This immediately gives rise to the question whether this feature is due to a self-reversal of magnetization or to a geomagnetic field-reversal.

The opaque mineralogy of specimens from the two groups is similar the only difference being the amounts and relative proportions of the minerals present in the specimens. For this reason, the opaque mineralogy of the specimens does not support the possibility of a self-reversal of magnetization.

Heating and cooling experiments on specimens from both groups always resulted in the specimens acquiring an artificial TRM parallel to the magnetic field in which they cooled. The same results were obtained by imparting an IRM to the specimens and at no time was any evidence whatsoever found to support a self-reversal or even partial self-reversal of magnetization.

With lack of evidence to the contrary, it must be concluded that the polarity difference between subzones B and C is due to a reversal of the earth's magnetic field which took place approximately 2000 m.y. ago.

SUMMARY AND CONCLUSIONS

All the fold tests carried out on palaeomagnetic data from the main zone in the eastern Bushveld Complex improved group statistics. This implies that the main zone acquired its present dip and dip direction after cooling to below its Curie temperature. From this, emplacement of the Bushveld magma must have been in the form of horizontal sheets. Upon cooling, crystallization took place to produce the igneous layering. At a later stage with the temperature below 578° isostatic readjustment of the crust commenced. This caused downbuckling of the crust and the associated subsidence of the Bushveld floor. Any model on the genesis and subsequent tectonic history of the Bushveld will have to be compatible with the above set of events.

The investigation succeeded in establishing magnetization directions for subzones B and C of the main zone, which are as follows:

Subzone B
group: BMZ2AFR: $N = 13$; $D = 10°$; $I = 64.7°$; $\alpha_{95} = 4.2°$; $k = 100$
with pole position: 17.3°N, 35.7°E
and polar error (dp, dm): 5.4° and 6.4° respectively

Subzone C
group: BMZ4AFR: $N = 3$; $D = 168.9°$; $I = -55.0°$; $\alpha_{95} = 20°$; $k = 37$
with pole position: 28°S, 161.7°W
and polar error (dp, dm): 20.1° and 28.6° respectively

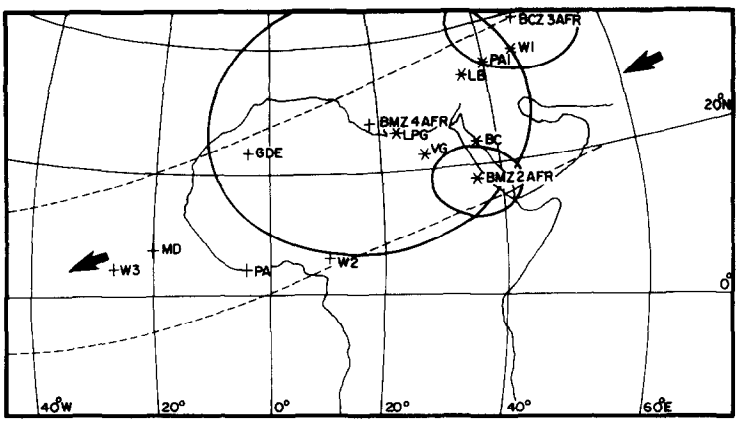


Fig. 21. APW path for Africa with selected pole positions shown for Southern Africa ca. 2070–1700 m.y. *PA1* (2067 m.y.)—Phalaborwa pyroxenite (Morgan and Briden, 1981); *W1–W3*—Waterberg sediments (stratigraphically arranged poles E, D and C in McElhinny, 1968); *VG* (1970 m.y.)—Vredefort Granophyre (Hargraves, 1970); *LPG* (2000 m.y.)—Limpopo gneisses (Morgan and Briden, 1981); *BC* (1950 m.y.)—Bushveld Complex (Gough and Van Niekerk, 1959); *LB* (1950 m.y.)—Losberg Gabbro Hargraves (1970); *GDE* (1935 m.y.)—Great Dyke extensions into Limpopo belt (Jones et al., 1975); *PA*—Phalaborwa syenite and dolerites (Morgan and Briden, 1981); *MD* (1850–1910 m.y.)—Mashonaland dolerites (McElhinny and Opdyke, 1964). Rock ages from Morgan and Briden (1981); *BMZ2AFR*—subzone B, main zone (this study); *BMZ4AFR*—subzone C, main zone (this study); *BCZ3AFR*—critical zone, Bushveld Complex (Hattingh, 1985). *—N poles; + —S poles.

The mixed magnetic polarity exhibited by site 11 at the base of the main zone suggests a polarity change which occur in subzone A of the main zone. This is confirmed by the polarity observed for the critical zone by Hattingh (1985). Whether this polarity change is due to a geomagnetic field reversal or a self-reversal of magnetization is at this stage unclear.

The second polarity change separate subzone B from subzone C. Subzone C exhibit a magnetization direction with a normal polarity (defined here as a magnetization direction in the same sense as the present-day magnetic field in southern Africa), and subzone B a reversed magnetic polarity. This polarity change is consistent with the overall polarity change which occur on the apparent polar wander path (APW) which indicates a geomagnetic field reversal c.a. 2000 m.y. ago (Fig. 21).

The palaeomagnetic pole positions for subzone B (pole BMZ2AFR) and subzone C (pole BMZ4AFR) are shown in Fig. 21, together with other pole positions from southern Africa. Unfortunately the ovals of 95% confidence around poles BMZ2AFR and BMZ4AFR overlap and consequently no conclusion on relative age differences between the two subzones can be made. However it is clear from Fig. 21 that there is an age difference between subzone B and the critical zone (pole BCZ; Hattingh, 1985). Subzone B had acquired its magnetization later than the critical zone, this implies a cooling of the mafic Bushveld magma from bottom to top.

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REFERENCES

Brandt, J.W., 1946. Corundum indicator, basic rocks and associated pegmatites in northern Transvaal. Trans. Geol. Soc. S. Afr., 49: 51–103.

Briden, J.C., 1972. A stability index of remanent magnetism. J. Geophys. Res., 77: 1401–1404.

Brock, A., 1971. An experimental study of palaeosecular variation. Geophys. J.R. Astron. Soc., 24: 303–317.

Cox, A., 1969. Confidence limits for the precision parameter κ. Geophys. J.R. Astron. Soc., 18: 545–549.

Creer, K.M. and Sanver, M., 1967. The use of the sun compass. In: D.W. Collinson, K.M. Creer and S.K. Runcorn (Editors), Methods in Palaeomagnetism. Develop. Solid Earth Geophys., 3. Elsevier, Amsterdam, pp. 11–15.

Dunlop, D.W., 1981. The rock magnetism of fine particles. Phys. Earth Planet. Inter. 26: 1–26.

Evans, M.E., McElhinny, M.W. and Gifford, A.C., 1968. Single domain magnetite and high coercivities in a gabbroic intrusion. Earth Planet. Sci. Lett., 4: 142–146.

Fleet, E.M., Bilcox, G.A. and Barnett, R.L., 1980. Oriented magnetite inclusions in pyroxenes from the Grenville province. Can. Min., 18: 89–99.

Gough, D.I., 1967. Notes on rock sampling for palaeomagnetic research. In: D.W. Collinson, K.M. Creer and S.K. Runcorn (Editors), Methods in Palaeomagnetism. Develop. Solid Earth Geophys., 3. Elsevier, Amsterdam, pp. 3–7.

Gough, D.I. and Van Niekerk, C.B., 1959. A study of the palaeomagnetism of the Bushveld gabbro. Philos. Mag., 4: 126–136.

Graham, J.W., 1949. The stability and significance of magnetism in sedimentary rocks. J. Geophys. Res., 54: 131–167.

Groeneveld, D., 1970. The structural features and the petrography of the Bushveld Complex in the vicinity of Stoffberg, eastern Transvaal. Geol. Soc. S. Afr., Spec. Publ., 1: 36–46.

Hargraves, R.B., 1970. Palaeomagnetic evidence relevant to the origin of the Vredefort Ring. J. Geol., 78: 253–263.

Hargraves, R.B. and Young, W.M., 1969. Source of stable remanent magnetism in Lambertville diabase. Am. J. Sci., 267: 1167–1177.

Hattingh, P.J., 1985. The palaeomagnetism of the Merensky Reef footwall rocks of the Bushveld Complex. Trans. Geol. Soc. S. Afr., in press.

Jones, D.L., Robertson, I.D.M. and McFadden, P.L., 1975. A palaeomagnetic study of Precambrian dyke swarms associated with the Great Dyke of Rhodesia. Trans. Geol. Soc. S. Afr., 78: 57–65.

McElhinny, M.W., 1964a. An improved method for demagnetising rocks in alternating magnetic fields. Geophys. J.R. Astron. Soc., 10: 369–374.

McElhinny, M.W., 1964b. Statistical significance of the fold test in palaeomagnetism. Geophys. J.R. Astron. Soc., 8: 338–340.

McElhinny, M.W., 1968. Palaeomagnetic directions and pole positions—IX. Geophys. J.R. Astron. Soc., 16: 207–224.

McElhinny, M.W. and Opdyke, N.D., 1964. The paleomagnetism of the Precambrian dolerites of eastern Southern Rhodesia an example of geologic correlation by rock magnetism. J. Geophys. Res., 69: 2465–2475.

Morgan, G.E. and Briden, J.C., 1981. Aspects of Precambrian palaeomagnetism, with new data from the Limpopo mobile belt and Kaapvaal craton in Southern Africa. Phys. Earth Planet. Inter., 24: 142–168.

Nagata, T., 1953. Rock-Magnetism. Maruzen Company, Tokyo, 225 pp.

Opdyke, N.D., 1972. Paleomagnetism of deep-sea cores. Rev. Geophys., 10: 213–249.

O'Reilly, W.O., 1984. Rock and mineral magnetism. Blackie and Son Ltd., Glasgow, 220 pp.

Scharlau, T.A., 1972. Petrographische und petrologische Untersuchungen in der Haupt-Zone des östlichen Bushveld-Komplexes, District Groblersdal/Transvaal. Diss., Johann Wolfgang Goethe Universität, Frankfurt/Main.

Soffel, H.C., 1971. The single-domain (multidomain) transition in intermediate titanonmagnetites. J. Geophys., 37: 451–470.

South African Committee for Stratigraphy, 1980. Stratigraphy of South Africa. Part 1. Lithostratigraphy of the Republic of South Africa, South West Africa/Namibia and the Republic of Bophuthatswana, Transkei and Venda. Geol. Surv. S. Afr., Hb. 8, 690 pp.

Watson, G.S.A., 1956. A test for randomness of directions. Mon. Not. R. Astron. Soc., Geophys. Suppl., 7: 153–159.

Willemse, J. and Viljoen, E.A., 1970. The fate of the argillaceous material in the gabbroic magma of the Bushveld Complex. Geol. Soc. S. Afr., Spec. Publ., 1: 336–366.